LABSHEET-4

Sum.js

```
function sum(a, b) {
  return a + b;
}
module.exports = sum;
```

Even-odd.js

```
function isEvenOrOdd(num) {
  return num % 2 === 0 ? "Even" : "Odd";
}
module.exports = isEvenOrOdd;
```

App1.js

```
const sum = require('./sum');
const isEvenOrOdd = require('./even-odd');
```

File-ops.js

```
const fs = require('fs');
// 1. Create student.txt and write info
fs.writeFile('student.txt', 'Name: Ashish Singh Rawat\nRoll No: 2024001\n', (err) => {
  if (err) throw err;
  console.log("student.txt created.");
  // 2. Append course
  fs.appendFile('student.txt', 'Course: Full Stack Web Development\n', (err) => {
    if (err) throw err;
    console.log("Course appended.");
    // 3. Read and print full content
    fs.readFile('student.txt', 'utf8', (err, data) => {
      if (err) throw err;
      console.log("\n--- File Content ---\n" + data);
      // 4. Rename to profile.txt
      fs.rename('student.txt', 'profile.txt', (err) => {
```

```
      if (err) throw err;
      console.log("File renamed to profile.txt");
    });
  });
});
```

Dir-ops.js

```
const fs = require('fs');
const path = require('path');
const dir = path.join(__dirname, 'uploads');
// 1. Create directory 'uploads'
fs.mkdir(dir, { recursive: true }, (err) => {
 if (err) throw err;
 // 2. Create 3 empty files inside uploads
 ['file1.txt', 'file2.txt', 'file3.txt'].forEach(f => fs.writeFileSync(path.join(dir, f), ''));
 console.log("Files created inside uploads folder.");
 // 3. List all files in the folder
 fs.readdir(dir, (err, files) => {
  if (err) throw err;
  console.log("Files in uploads:", files);
  // 4. Delete one file
  fs.unlink(path.join(dir, 'file2.txt'), (err) => {
   if (err) throw err;
   console.log("file2.txt deleted.");
  });
 });
});
```

Path-demo.js

```
const path = require('path');
const samplePath = '/home/user/data/report.pdf';
```

```
console.log("Directory Name:", path.dirname(samplePath));
console.log("Base Name:    ", path.basename(samplePath));
console.log("Extension:    ", path.extname(samplePath));
console.log("Resolved Path: ", path.resolve('relative/file.txt'));
```

Sys-report.js

```
const os = require('os');
console.log("Platform:      ", os.platform());
console.log("Architecture:   ", os.arch());
console.log("Total Memory:   ", (os.totalmem() / (1024 ** 3)).toFixed(2), "GB");
console.log("Free Memory:    ", (os.freemem() / (1024 ** 3)).toFixed(2), "GB");
console.log("CPU Cores:      ", os.cpus().length);
console.log("System Uptime:  ", (os.uptime() / 60).toFixed(2), "minutes");
```

Order-events.js

```
const { EventEmitter } = require('events');
const orderEmitter = new EventEmitter();
// Define listener
orderEmitter.on('orderPlaced', (order) => {
  console.log("----- Receipt -----");
  console.log(` Order ID : ${order.id}` );
  console.log(` Item     : ${order.item}` );
  console.log(` Amount   : $${order.amount}` );
  console.log("-------------------\n");
});
// Emit event 3 times
orderEmitter.emit('orderPlaced', { id: 101, item: 'Laptop Stand', amount: 35 });
orderEmitter.emit('orderPlaced', { id: 102, item: 'Mechanical Keyboard', amount: 80 });
orderEmitter.emit('orderPlaced', { id: 103, item: 'USB-C Cable', amount: 12 });
```

Server.js

```
const http = require('http');
const url = require('url');
const students = [
```

```
  { id: 1, name: "Alice", course: "CSE" },
  { id: 2, name: "Bob", course: "ECE" },
  { id: 3, name: "Charlie", course: "IT" }
];
const server = http.createServer((req, res) => {
  const parsedUrl = url.parse(req.url, true);
  const pathname = parsedUrl.pathname;
  if (req.method === 'GET') {
    // Route 1: Welcome message
    if (pathname === '/') {
      res.writeHead(200, { 'Content-Type': 'text/plain' });
      return res.end("Welcome to the Student API!");
    }
    // Route 2: Hardcoded JSON array of 3 students
    if (pathname === '/students') {
      res.writeHead(200, { 'Content-Type': 'application/json' });
      return res.end(JSON.stringify(students));
    }
    // Route 3: Dynamic route /students/:id
    const studentMatch = pathname.match(/^\/students\/(\d+)$/);
    if (studentMatch) {
      const studentId = parseInt(studentMatch[1], 10);
      const student = students.find(s => s.id === studentId);
      if (student) {
        res.writeHead(200, { 'Content-Type': 'application/json' });
        return res.end(JSON.stringify(student));
      } else {
        res.writeHead(404, { 'Content-Type': 'application/json' });
        return res.end(JSON.stringify({ error: "Student not found" }));
      }
    }
```

```
    // Task 8: /search query parameter parsing
   if (pathname === '/search') {
     const keyword = parsedUrl.query.keyword || 'None';
     res.writeHead(200, { 'Content-Type': 'application/json' });
     return res.end(JSON.stringify({ queryKeyword: keyword }));
   }
 }
 // Any other route: 404
 res.writeHead(404, { 'Content-Type': 'text/plain' });
 res.end("404 Not Found");
});
server.listen(3000, () => console.log("Server running on http://localhost:3000"));
```

Stream-demo.js

```
const fs = require('fs');
const filename = 'large.txt';
// Create file with 60 lines
const lines = Array.from({ length: 60 }, (_, i) => `Line ${i + 1}: Sample stream text data\n`).join('');
fs.writeFileSync(filename, lines);
// Read using stream with custom buffer size
const readStream = fs.createReadStream(filename, { highWaterMark: 128 });
readStream.on('data', (chunk) => {
  console.log(`Received chunk size: ${chunk.length} bytes`);
});
readStream.on('end', () => {
  console.log("Finished reading stream.");
});
```

Id-gen.js

```
const { nanoid } = require('nanoid');
console.log("Generated 5 Unique IDs:");
for (let i = 0; i < 5; i++) {
  console.log(`${i + 1}:`, nanoid());
```

```
}
```

calc-cli.js

```
const args = process.argv.slice(2);
if (args.length < 3) {
  console.log("Usage: node calc-cli.js <num1> <num2> <operator>");
  process.exit(1);
}
const num1 = parseFloat(args[0]);
const num2 = parseFloat(args[1]);
const op = args[2];
let result;
switch (op) {
  case '+': result = num1 + num2; break;
  case '-': result = num1 - num2; break;
  case '*': result = num1 * num2; break;
  case '/': result = num2 !== 0 ? num1 / num2 : 'Error: Division by zero'; break;
  default: result = 'Invalid operator';
}
console.log(`Result: ${result}`);
```

env-demo.js

```
const http = require('http');
// Read PORT from environment or default to 4000
const PORT = process.env.PORT || 4000;
const server = http.createServer((req, res) => {
  res.writeHead(200, { 'Content-Type': 'text/plain' });
  res.end(`Server running on port ${PORT}`);
});
server.listen(PORT, () => {
 console.log(`Server started and listening on port: ${PORT}`);
});
```

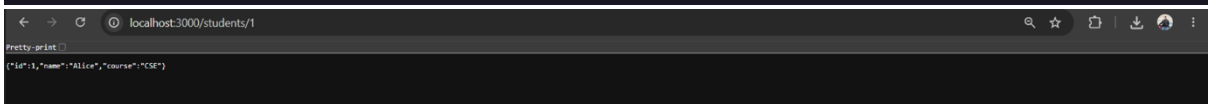
```
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node app1.js
Sum (10 + 25): 35
Check 14: Even
Check 7: Odd
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node file-ops.js
student.txt created.
Course appended.

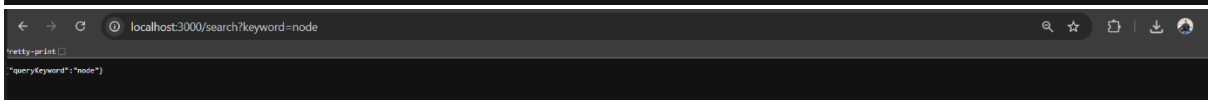
--- File Content ---
Name: Ashish Singh Rawat
Roll No: 2024001
Course: Full Stack Web Development

File renamed to profile.txt
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node dir-ops.js
Files created inside uploads.
Files in uploads: [ 'file1.txt', 'file2.txt', 'file3.txt' ]
file2.txt deleted.
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node path-demo.js
Directory Name: /home/user/data
Base Name:      report.pdf
Extension:      .pdf
Resolved Path:  C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4\relative\file.txt
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node sys-report.js
Platform:       win32
Architecture:   x64
Total Memory:   15.31 GB
Free Memory:    1.94 GB
CPU Cores:      16
System Uptime:  11173.04 minutes
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node order-events.js
----- Receipt -----
Order ID : 101
Item     : Laptop Stand
Amount   : $35
-------------------

----- Receipt -----
Order ID : 102
Item     : Mechanical Keyboard
Amount   : $80
-------------------

----- Receipt -----
Order ID : 103
Item     : USB-C Cable
Amount   : $12
-------------------
```







```
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node stream-demo.js
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 128 bytes
Received chunk size: 51 bytes
Finished reading stream.
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node id-gen.js
Generated 5 IDs:
1: 1gza4DAiTKvbE-ZC1VGbm
2: BCJZO2sSC_v6Qtl5J4L0Q
3: sco443qrQGrFrPAKIIaOJ
4: euhfweDD-2fjqPBLKvEx5
5: quxfJ82MTLwjZG6CdyEeE
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node calc-cli.js 10 5 +
Result: 15
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> node calc-cli.js 20 4 /
Result: 5
PS C:\Users\rawat\Desktop\Full Stack Clg lab\labsheet4> $env:PORT="5000"; node env-demo.js
Server started and listening on port: 5000
```

